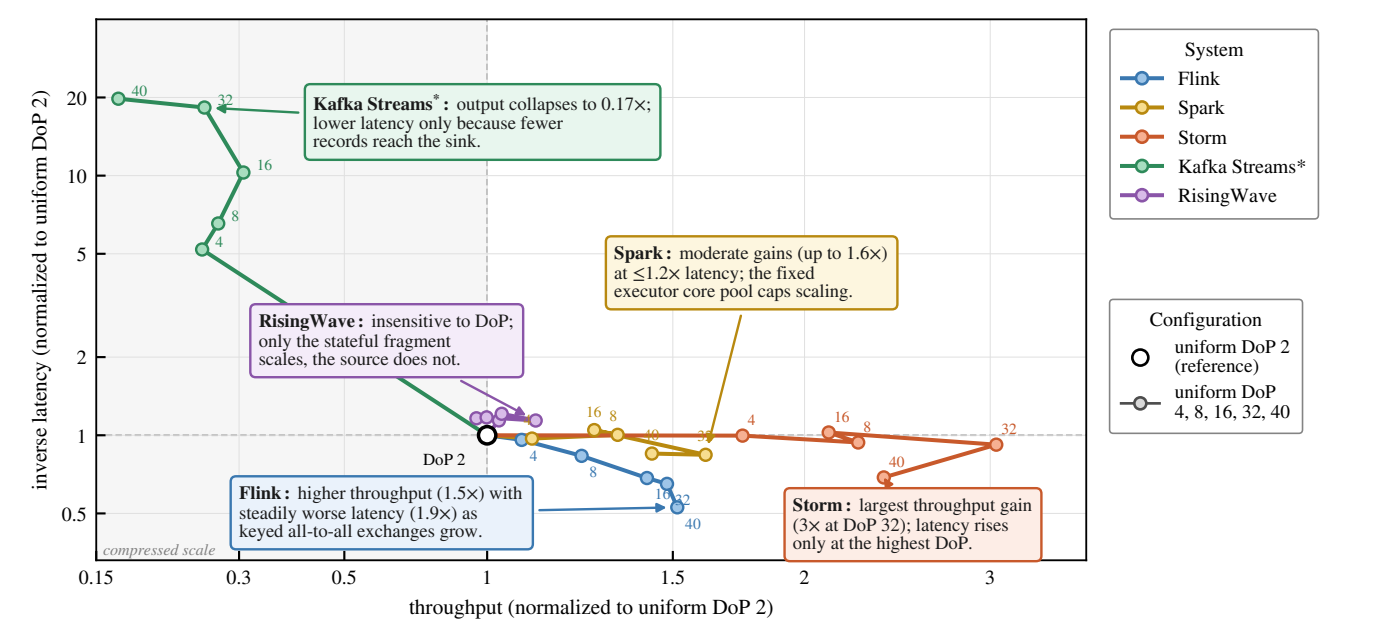
inverse latency (normalized to uniform DoP 2)
throughput (normalized to uniform DoP 2)
20
10
5
2
1
0.5
0.15
0.3
0.5
1
1.5
2
3
compressed scale
DoP 2
Kafka Streams*: output collapses to 0.17×; lower latency only because fewer records reach the sink.
RisingWave: insensitive to DoP; only the stateful fragment scales, the source does not.
Spark: moderate gains (up to 1.6×) at ≤1.2× latency; the fixed executor core pool caps scaling.
Flink: higher throughput (1.5×) with steadily worse latency (1.9×) as keyed all-to-all exchanges grow.
Storm: largest throughput gain (3× at DoP 32); latency rises only at the highest DoP.
System
Flink
Spark
Storm
Kafka Streams*
RisingWave
Configuration
uniform DoP 2 (reference)
uniform DoP 4, 8, 16, 32, 40